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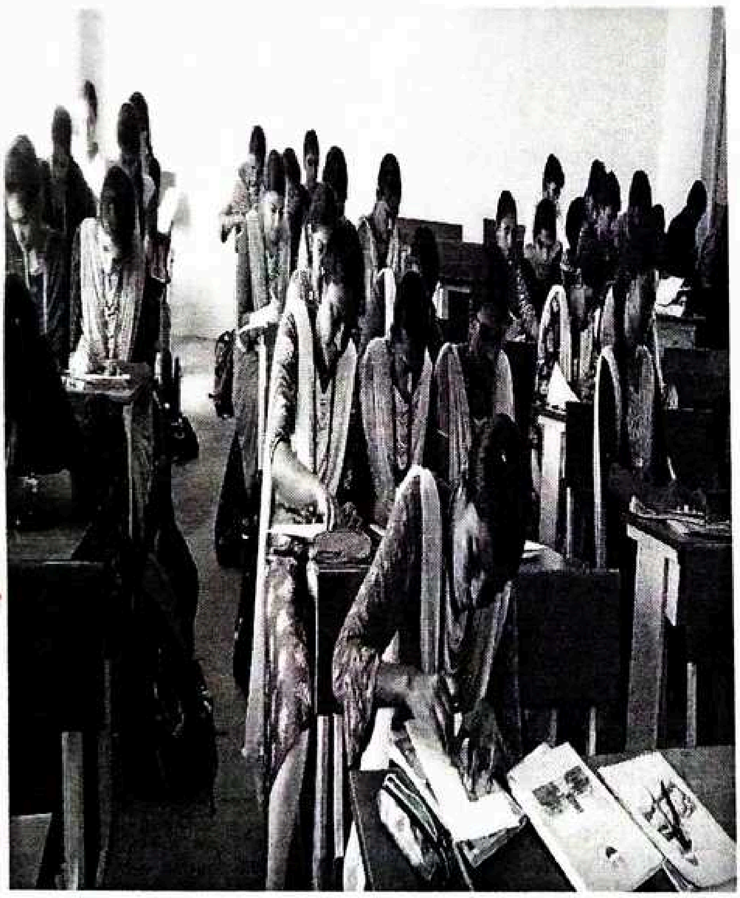
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

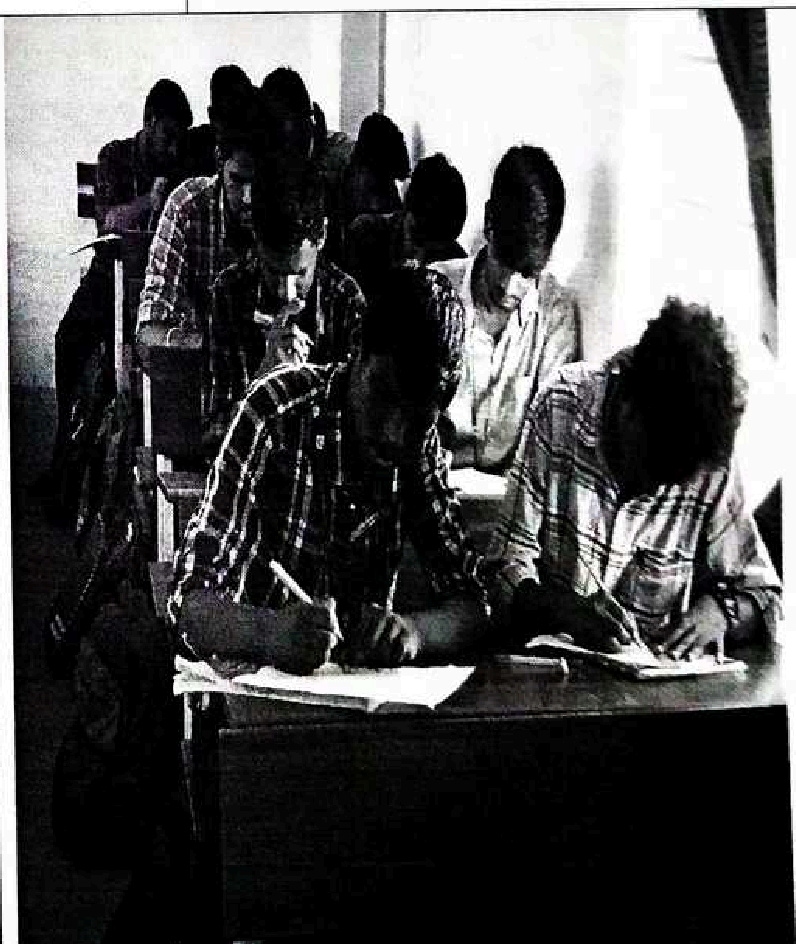
Department of Electronics and Communication Engineering
Academic Year: 2022 – 2023(Even Semester)

INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : IV Semester B.E. ECE A
Course Code & Title : EC3451 & Linear Integrated Circuits
Name of the Faculty member: Mrs.G.Subhashini

Sl.No.	Date	Topic(s)	Activity*	Reference
UNIT I BASICS OF OPERATIONAL AMPLIFIERS				
1.	10.02.2023	Voltage Sources, Voltage References – Band gap	One Minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/





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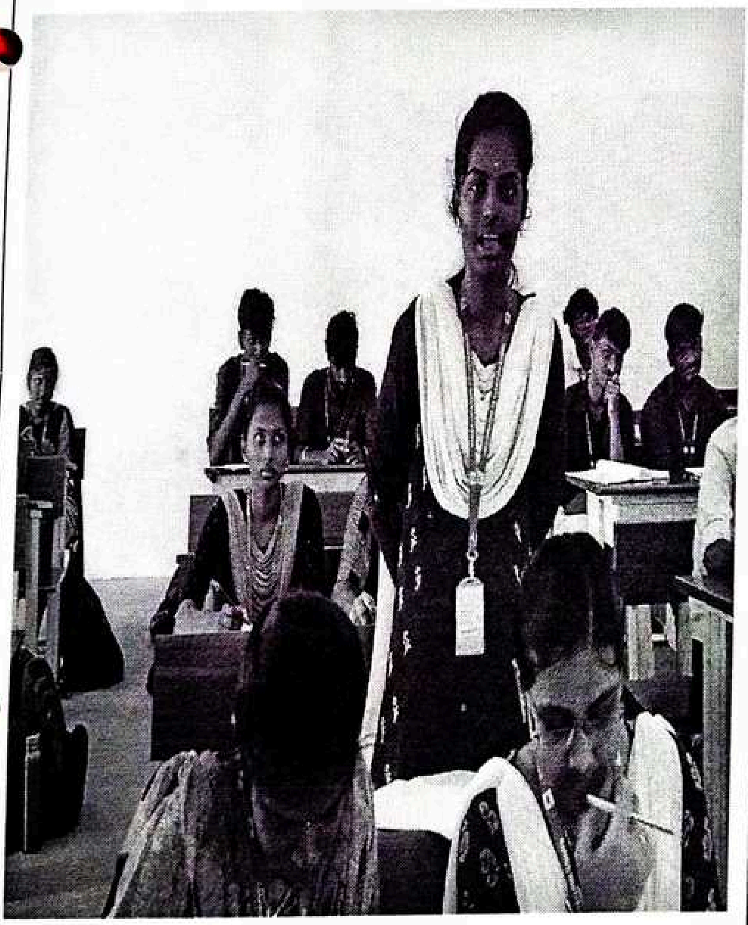
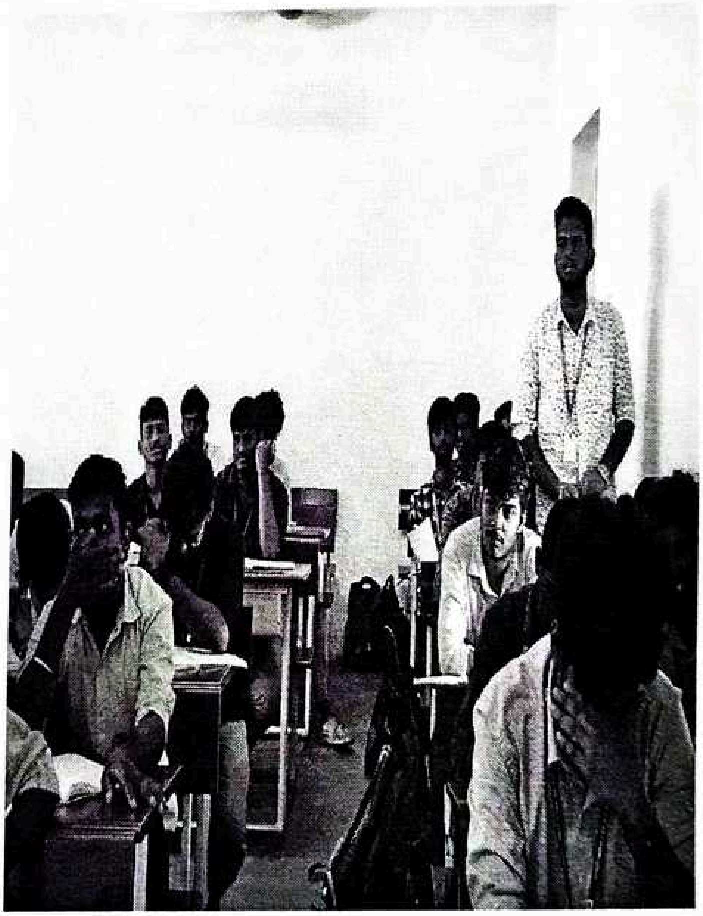
UNIT II - Applications of Operational Amplifiers				
1.	07.03.2023	Precision rectifier, Peak detector, Clipper and Clamper	Head and Talk	https://gdhr.wa.gov.au/learning/teaching-strategies/finding-out/head-talk





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ACTIVITY: Flipped Classroom
Lesson Plan

Key elements of Flipped Classroom	• To provide an opportunity for students to gain first exposure to content prior to class. • To provide an incentive for students to prepare for class. • To provide in-class activities that focus on higher-level cognitive activities. • To provide a mechanism to assess student understanding.
Topic	Application of Multipliers
Justification for Choosing the Topic using Flipped Class	The topic was chosen for flipped class is there are many applications for Multipliers. The Unit 3 deals with the details of Analog multiplier ICs and its applications and functionalities, various techniques used for multipliers and their operations. Since the students are exposed to the details of Multiplier, I taught of giving the topic ***'Application of Multiplier'*** for flipped class. This topic can be learnt in self-learning mode. ❖ As there are many applications of Multiplier, only 6 applications in various domains were discussed in their prescribed text book. If this flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related to specific application. Thus 6 applications of Multiplier will be covered in 2 periods through this activity which is not possible in traditional teaching. This is the reason for choosing this topic for flipped class activity.
Subject	Linear Integrated Circuits

Time Needed	At Home: 10 minutes to watch each video, total of 20 minutes Students will be assigned videos two days in advance to give them timeto watch them at home and complete At-Home Activities In class: 50 minutes class period
Learning Objective	To explain the application of Multiplier in various fields like Digital Signal Processing, VLSI, Communication and Embedded system
Expected Outcome	At the end of the flipped class activity, ❖ Students will be exposed to collaborative learning environment. ❖ The students will develop better understanding of the conceptsand principles. ❖ The students will be able to explain the various applications of Multipliers with realtime examples in the areas of Digital Signal Processing, VLSI, Communication and Embedded system. ❖ The problem solving skills of the students will be improved. ❖ It will help students to perform better at end semester examination.
Student Learning Resources(Home) ie: videos, website, teachercreated materials	NPTEL video #1: Analog Multipliers Professor: Prof.K.Radhakrishna Rao, Lecture series:19 Course name:Analog ICs https://www.digimat.in/nptel/courses/video/108106068/L19.html The NPTEL video discusses about Multiplier in detail and shows the real time application with demonstration. The **NPTEL video** gives the detailed presentation of Multipliers. Kindly make use of book chapters for preparing the presentations and apart from the details any other information regarding the applications can also be added from other online resources with proper citations. Prepare a presentation for 7 minutes containing the ✓ Introduction ✓ How Analog ICs play a role in the application related to your group ✓ Block diagram

	✓ Technical Information's ✓ Type of Multiplier used ✓ Advantages and Disadvantages ❖ The book chapters will be shared to the students as materials and video like from NPTEL will be posted related application of Multipliers in the course website. ❖ The learning material of book chapters consist of details for 6 applications such as 1. Voltage Divider using multiplier 2. Square rooting circuit using multiplier 3. Frequency doubler using multiplier 4. Phase angle detection using multiplier 5. Rectifier using multiplier 6. Voltage controlled attenuator
Student Learning Activities(Home)	1. Students will watch 2 YouTube videos about the Multipliers. 2. While watching the videos, students will note the important information on the specified topic to help them collect information and write questions that they still have. 3. Reading of learning material: 10 days time duration 4. Preparation of presentation: 4 days time duration 5. Discussions and clarification of doubts: Any time within the 2 weeks time for discussionwithin the group mates and any clarification regarding the flipped class
Classroom Activities and Materials	1. Presenting the topic assigned by each group – 5 minutes duration 2. Question & Answering and Oral feedback: Class will open with a question and answer session and the teacher will ask questions raised by students if noone provides questions in class. These will serve as the discussion.– 2 minutes duration

Formation of the groups and Justification:

The formation of group for flipped class will be given to student's choice.

Number of Applications (topics): 06

Number of groups: 9

Class Strength: 63

Members per group: 7

The choice of the group type is heterogeneous grouping, the condition of the grouping will informed to the students. ie. A group should be mixer of advance learners, average learners and slow learners. The students group was given the students choices with the above said conditions because the students should learn about applications of Multiplier and prepare a presentation about the topic. Since a detailed presentation should be made by each group, if the choice is given to the student's only successful learning will happen. The list of group is shown in the evaluation table for flipped class given below.

Implementation process:

- The students were informed about the flipped class activity atleast 2 weeks before the implementation of the activity.
- The materials like documents, videos related to the topic will be posted in the course website - Canvas.
- The grouping of students will also be done well in advance and each group is allotted with one topic of application of their choice among the 6 applications listed. (2 weeks before implementation)
- An announcement will be posted once in two days about the activity as reminder.
- A Discussion forum will be opened for flipped class, in which the group members can use for discussion outside class. Using the 'your questions about flipped class' – doubts of the students can be posted and the instructor will clarify within the 8 hours of post.
- A whatsapp group will be created for flipped class activity to clarify the doubts and discussions.
- The students have to prepare a presentation for 5 minutes with the concept learnt and 2 minutes for Question & answering and tell about the contribution of each member.
- The inside class activity: presentation will be planned for 1 period. In which each group has to present their topic chosen and others will also listen to the presentation.
- Time management: The flipped class activity should be able to complete in the planned duration.
- The applications of Multiplier are covered to the whole class and the handouts prepared on the topic will be shared within the group members.
- Each group will be assessed through the content of presentation, communication skills, Q&

A, contribution by each member as shown in the table below. Both formative and summative assessment will be made.

- ✓ Content of presentation (20 Marks)
- ✓ Communication skills (10 Marks)
- ✓ Q & A(10 Marks)
- ✓ Contribution (10 Marks)

- ❖ The main objective of the flipped class activity is each group will learn one application and presentation helps the other members also to know about all the application topics.
- ❖ Thus, 6 applications of Multiplier will be covered in 1 period throughthis activity.
- ❖ After completion of presentation, both oral and written feedbacks will be collected about the flipped class from the students.
- ❖ The evaluation on the flipped class is done finally and will be assessed based on the following parameters. The marks will be considered for final internal mark calculation.

Student Feedback to access the success of the flipped class:

The feedback about the success of the flipped class will be collected in two modes

- ❖ **Oral feedback**
- ❖ **Written Feedback**

Oral Feedback:
The oral feedback will be collected at the end of the presentation by the students about

- ✓ Experience in doing this flipped class
- ✓ Contribution by each member of the group.
- ✓ What was the learning outcome of such activity?
- ✓ What is the difference you felt between traditional teaching and flipped class?

Written Feedback:

A written feedback will be collected with the following questionnaire at the end of completing the activity.

Students Feedback: Flipped Class Room Activity

1. The Flipped classroom is more engaging than traditional classroom instruction.?

   ☐ Disagree ☐ Agree ☐ Neither agree or disagree

2. I like watching the lessons on video.?

   ☐ Disagree ☐ Agree ☐ Neither agree or disagree

3. Learning how to use a Flipped Classroom will benefit me in my future education.

☐ Disagree ☐ Agree ☐ Neither agree or disagree

4. The Flipped Classroom gives me greater opportunities to communicate with other students.

   ☐ Disagree ☐ Agree ☐ Neither agree or disagree

Name & Signature of the Student with date

Based on the feedback collected, the success of the activity can be assessed and from the challenges faced during preparation the problems can be identified and rectified in the future flipped class activities.

HoD

Signature of the Faculty

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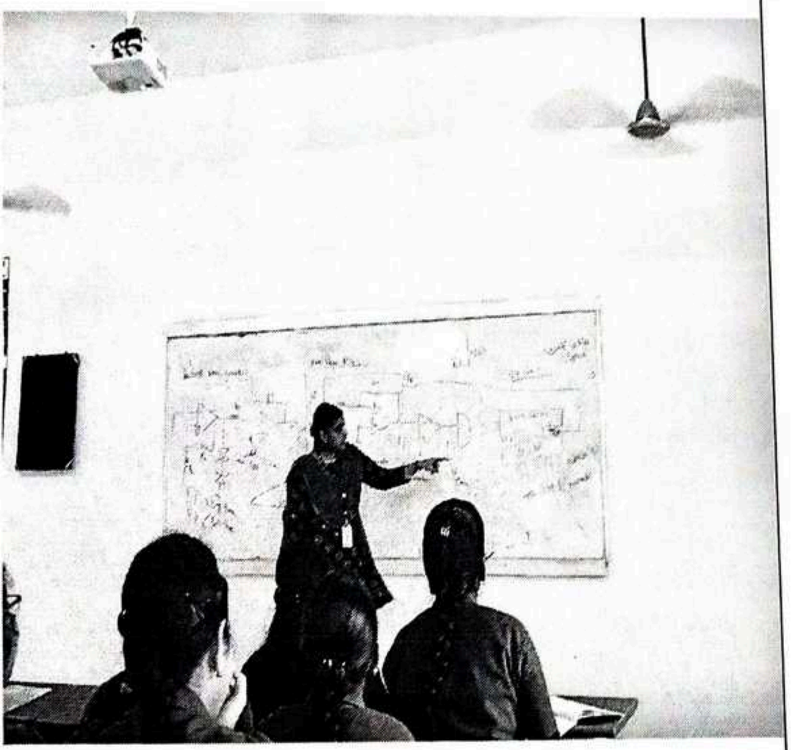
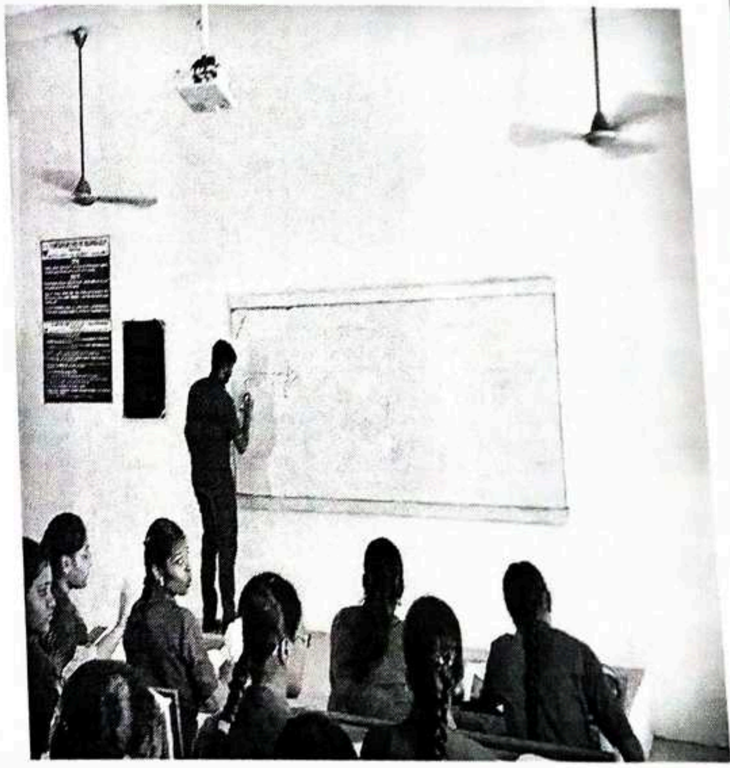
Sl.No.	Date	Topic(s)	Activity*	Reference
UNIT III – ANALOG MULTIPLIER AND PLL				
1.	20.03.2023	Analog Multiplier ICs and their applications	Flipped Class	https://www.panopto.com/blog/what-is-a-flipped-classroom/





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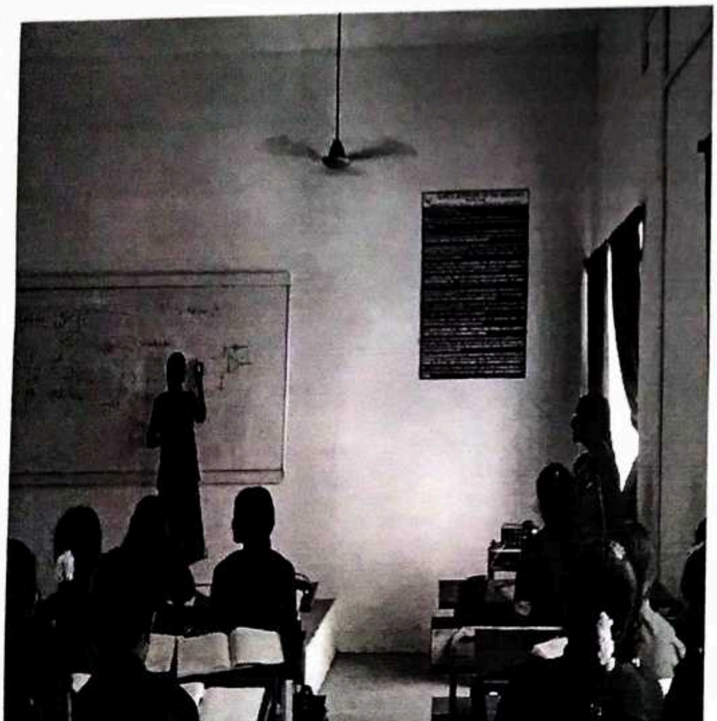
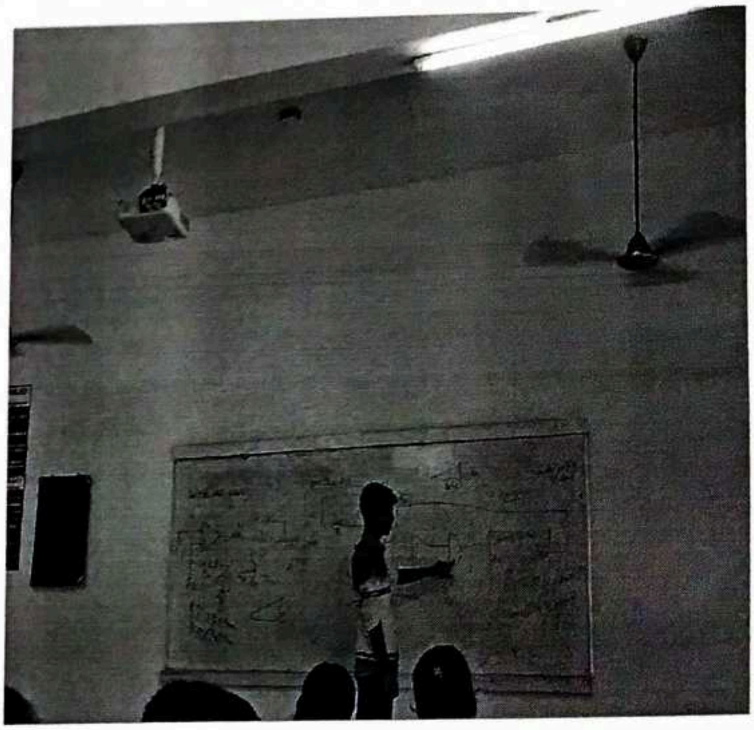
UNIT IV - ANALOG TO DIGITAL AND DIGITAL TO ANALOG CONVERTERS				
1.	19.04.2023	A/D Converter - Specifications	Role Play	https://www.teachingenglish.org.uk/professional-development/teachers/planning-lessons-and-courses/articles/role-play





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Name of the Faculty member: Mrs.G.Subhashini
Topic: Triangular and saw tooth wave generators

Learning Outcomes:
The students will be able to

- Represent the waveform generators using opamp circuits
- Differentiate Triangular and saw tooth wave generators
- Participate actively in the learning process.
- Discuss the concepts, convey ideas, share the views to really understand the Triangular and saw tooth wave generators
- Communicate effectively by sharing their views among the teammates.

Justification for choosing the topic for the activity:

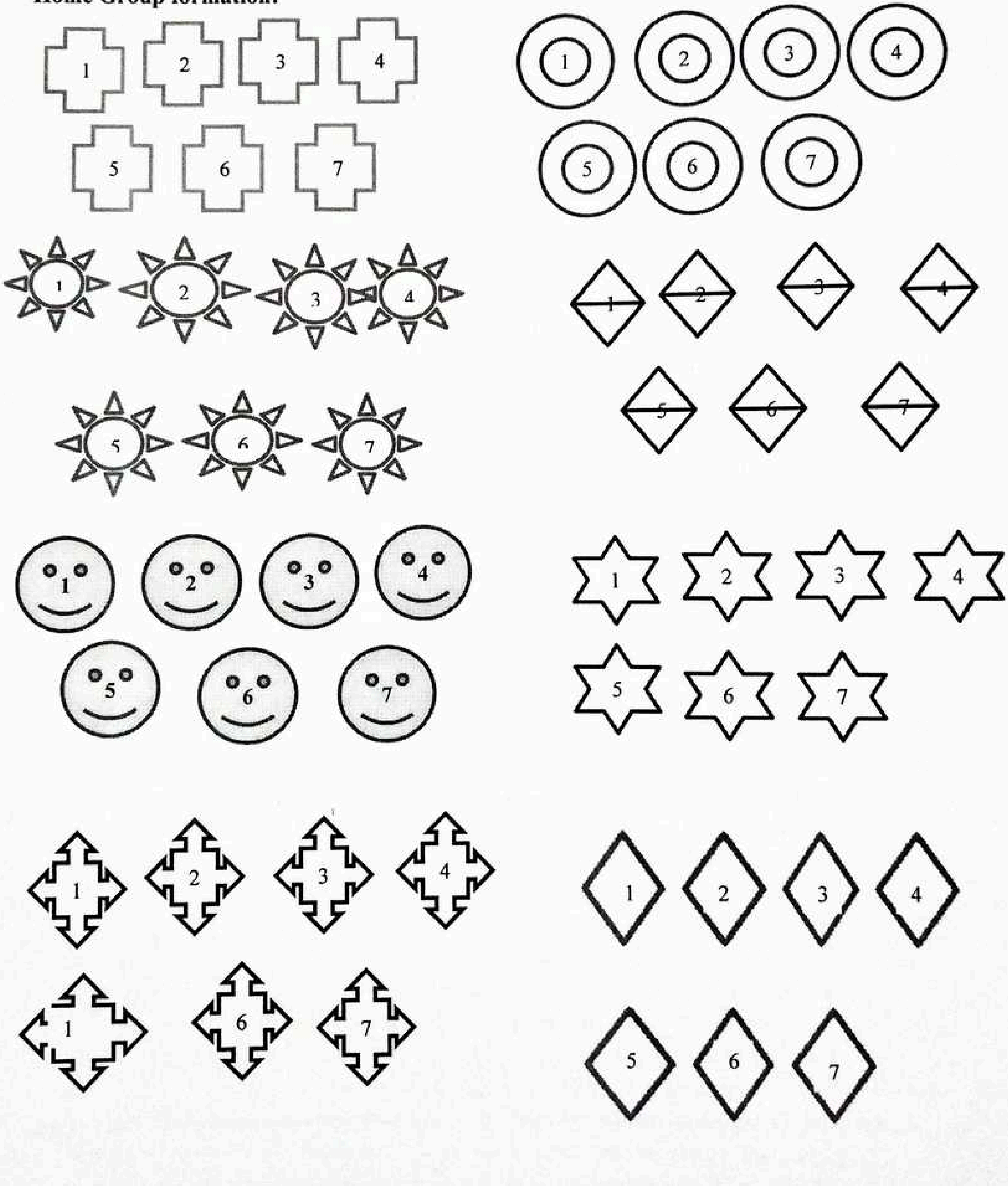
The concept of Waveform – Triangular wave, Sawtooth wave and Square wave are already learnt in Electronic circuits, Signals and systems as well as in the Circuits and simulation Lab sessions. This topic is related to the same concepts and easy to understand. The activity **'JIGSAW'** is used to recall the concepts learnt and enhance the learning through this collaboration.

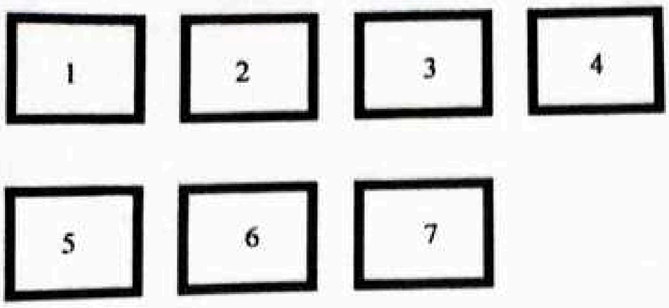
Plan for Implementation:

- The material related to the topic, video will be posted 1 week before the conduct of activity.
- **Activity:** JIGSAW- The Class Strength is divided into 9 Shapes Groups with 7 members and 7th (last group) consists of 9 members as shown below using Random Lot System.
- Each group is allotted with a shape and its known as **Home group** as shown below.
- 7 Shapes are named 1,2,3,4,5,6,7,. All the 1's in different Shapes are made to form Expert group.
- Each Expert group is allotted with one topic
  - ✓ Wave form generators Introduction
  - ✓ Types of Wave form generators
  - ✓ Triangular wave generators
  - ✓ Saw tooth wave generators
  - ✓ Working of Triangular wave generators
  - ✓ Working of Sawtooth wave generators
  - ✓ Differentiate Triangular and saw tooth wave generators
- The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas.
- The Expert group members should go to their original shape group and discuss the points (30 Minutes) to the other members.

- ❖ All the members in home group learnt about all the topics through expert group members.
- ❖ Finally one from each shape group should summarize the points of the topic Triangular and saw tooth wave generators to the class (15 Minutes).
- ❖ Review of points and conclusion of the activity (5 minutes)
- ❖ Thus the concept of Triangular and Sawtooth wave generators is known to all by involving the students actively.

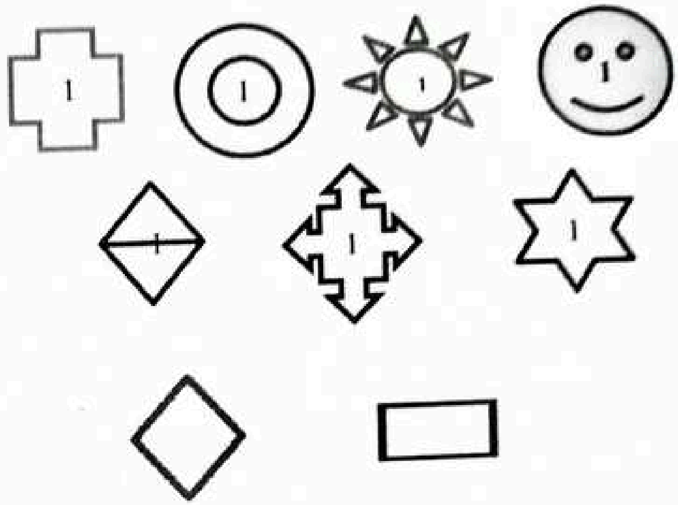
Home Group formation:

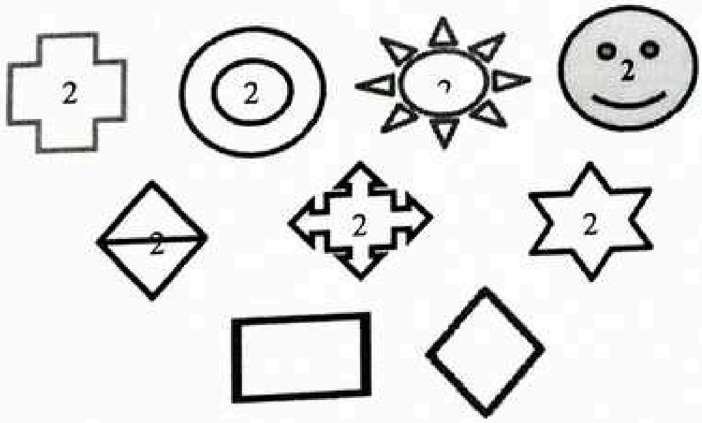




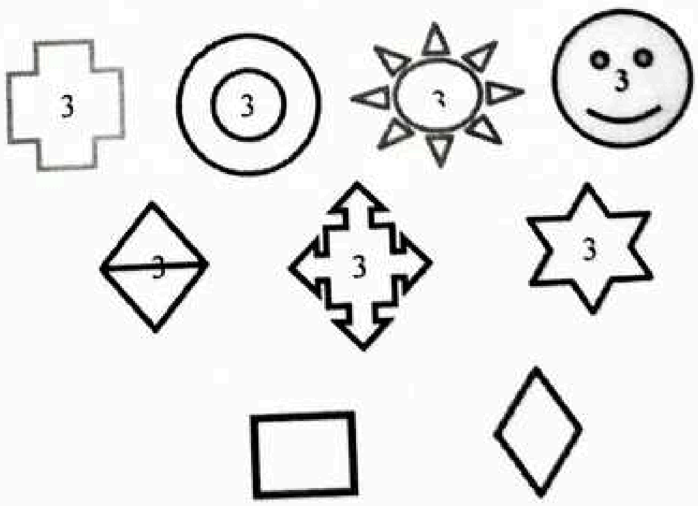
❖ Expert group consist of members from each shape and assigned with a topic for discussion (10 Minutes) which has been already learnt in III Semester subject.

GROUP 1
Topic: Waveform generators Introduction

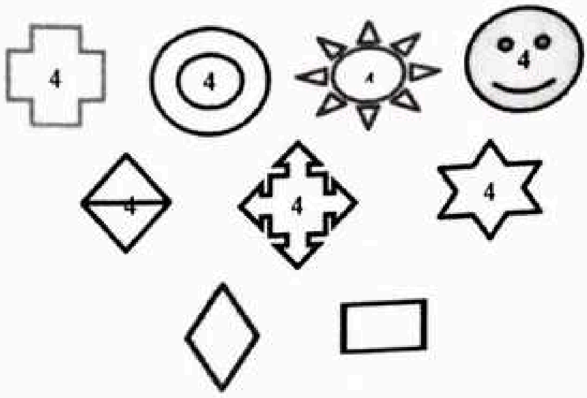




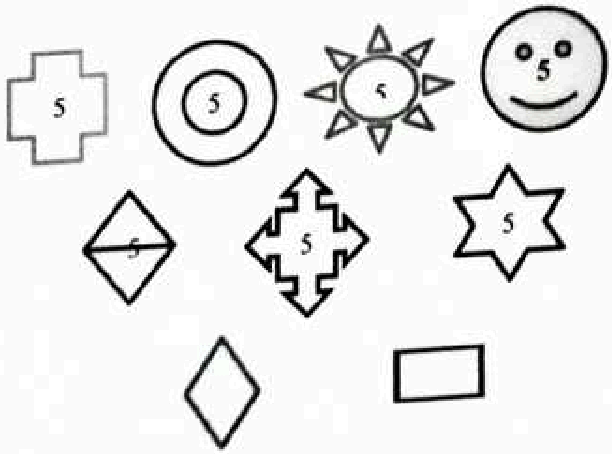
GROUP 2
Topic: Types of Wave form generators



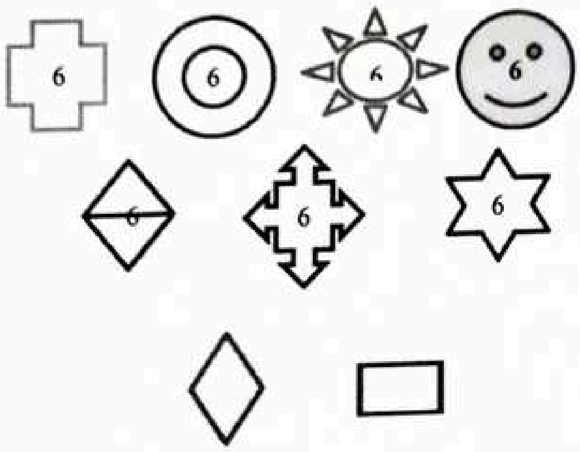
GROUP 3
Topic: Triangular wave generators



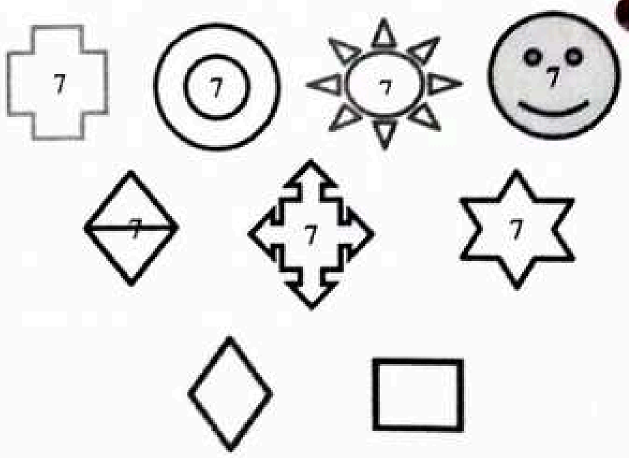
GROUP 4
Topic: Sawtooth wave generators



GROUP 5
Topic: Working of Triangular wave generators



GROUP 6
Topic: Working of Triangular wave generators



GROUP 7
Differentiate Triangular and sawtooth wave generators

References:

Kindly refer the following materials before coming to the Class

- https://www.jigsaw.org/
- https://www.newcastle.edu.au/data/assets/pdf_file/0016/109600/Jigsaw-learning-activity.pdf
- D.RoyChoudhry, Shail Jain, "Linear Integrated Circuits", New Age International Pvt. Ltd., 2018, Fifth Edition. (Page No: 220)

8/6/2023.